AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF COMPUTING CHENNAI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #2

Activity Name: GPIO Switch Interfacing

Name of Student:	**Chetan Kumar G**
Registration No.:	**CH.SC.U4CSE24109**
Activity Date:	**28|07|2026**
Submission Date:	**28|07|2026**

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	**10**	

Faculty In-charge

Write an embedded C program for ARM architecture to interface switches. Debug and verify the program using IDE on a development computer system.

PROGRAM:

```
//CH.SC.U4CSE24109
#include <LPC21xx.h>
#define SW1  (1<<0)      // Switch at P0.0
#define SW2  (1<<1)      // Switch at P0.1
#define LED1 (1<<16)     // LED1 at P0.16
#define LED2 (1<<17)     // LED2 at P0.17

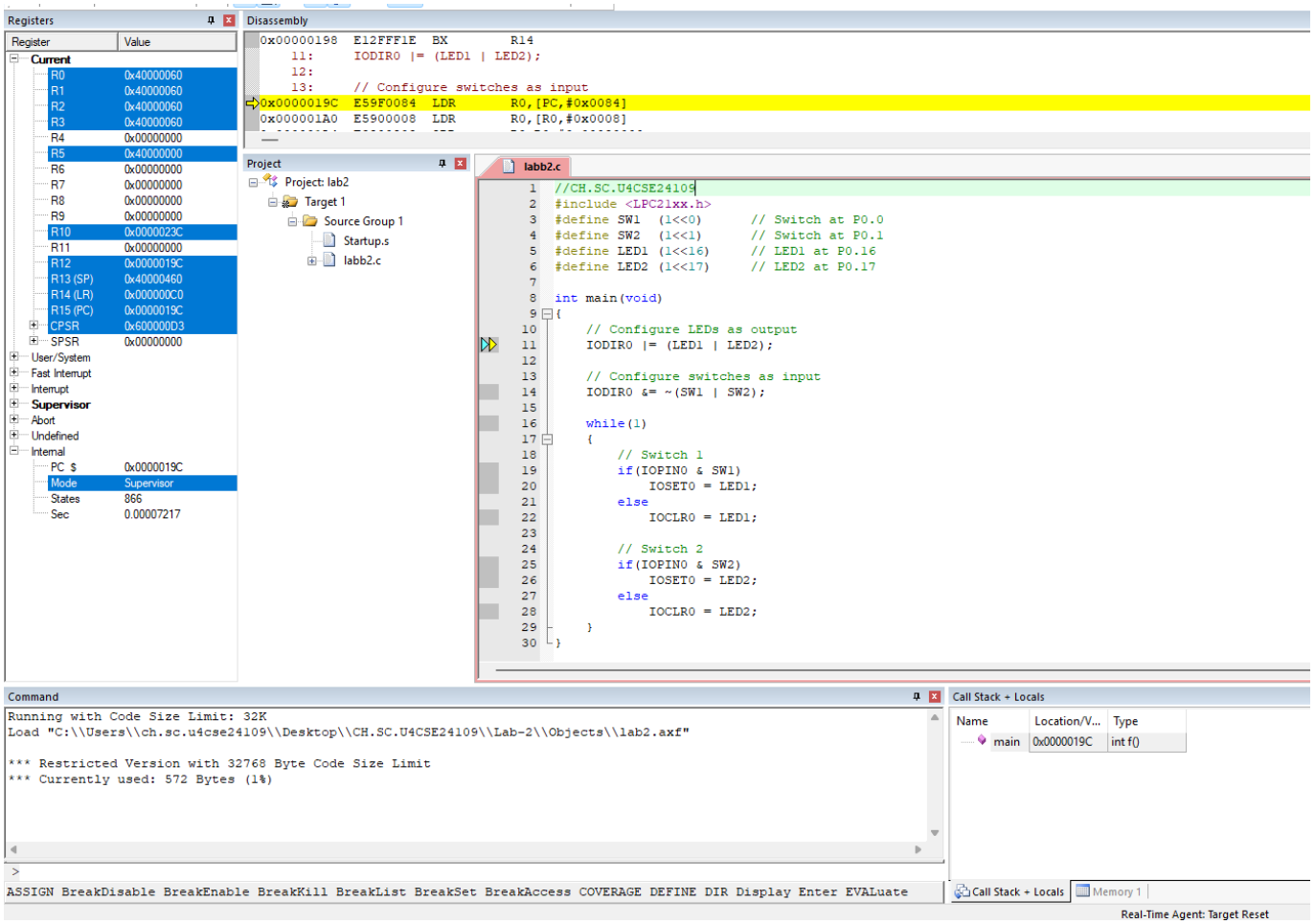
int main(void)
{
    // Configure LEDs as output
    IODIR0 |= (LED1 | LED2);

    // Configure switches as input
    IODIR0 &= ~(SW1 | SW2);

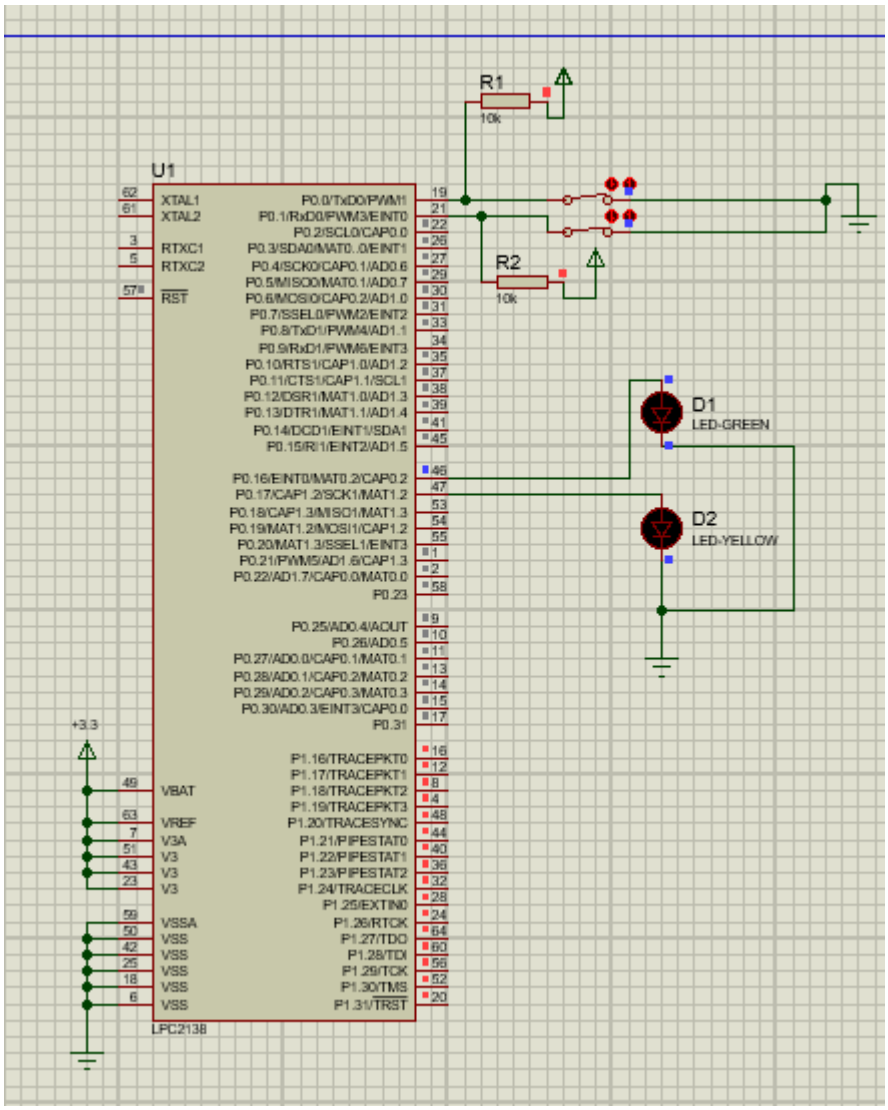
    while(1)
    {
        // Switch 1
        if(IOPIN0 & SW1)
            IOSET0 = LED1;
        else
            IOCLR0 = LED1;

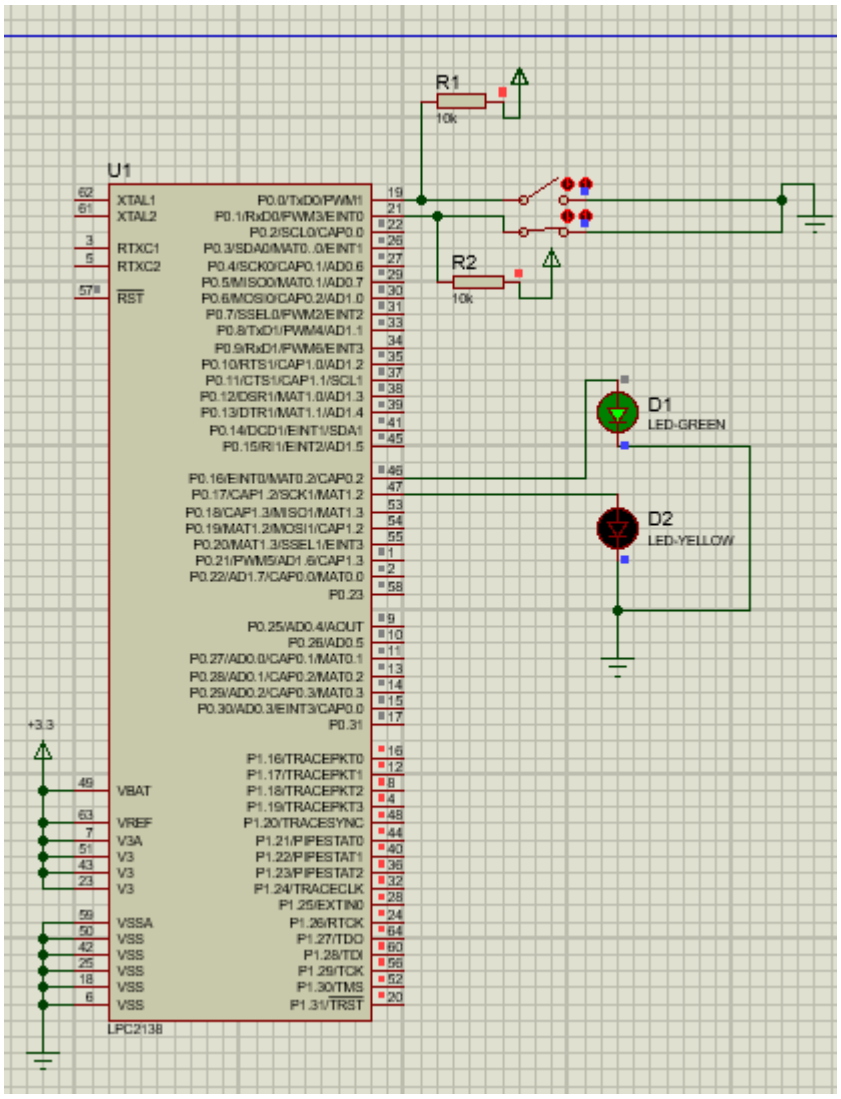
        // Switch 2
        if(IOPIN0 & SW2)
            IOSET0 = LED2;
        else
            IOCLR0 = LED2;
    }
}
```

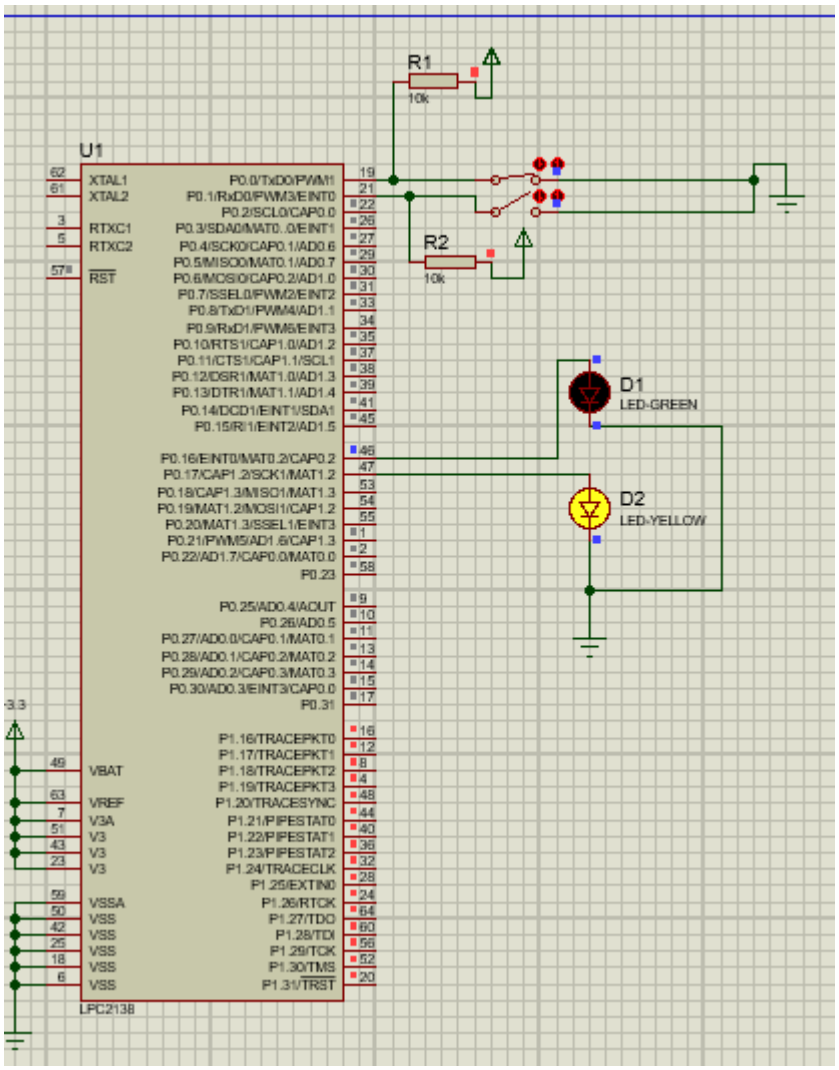
OUTPUT WINDOW:

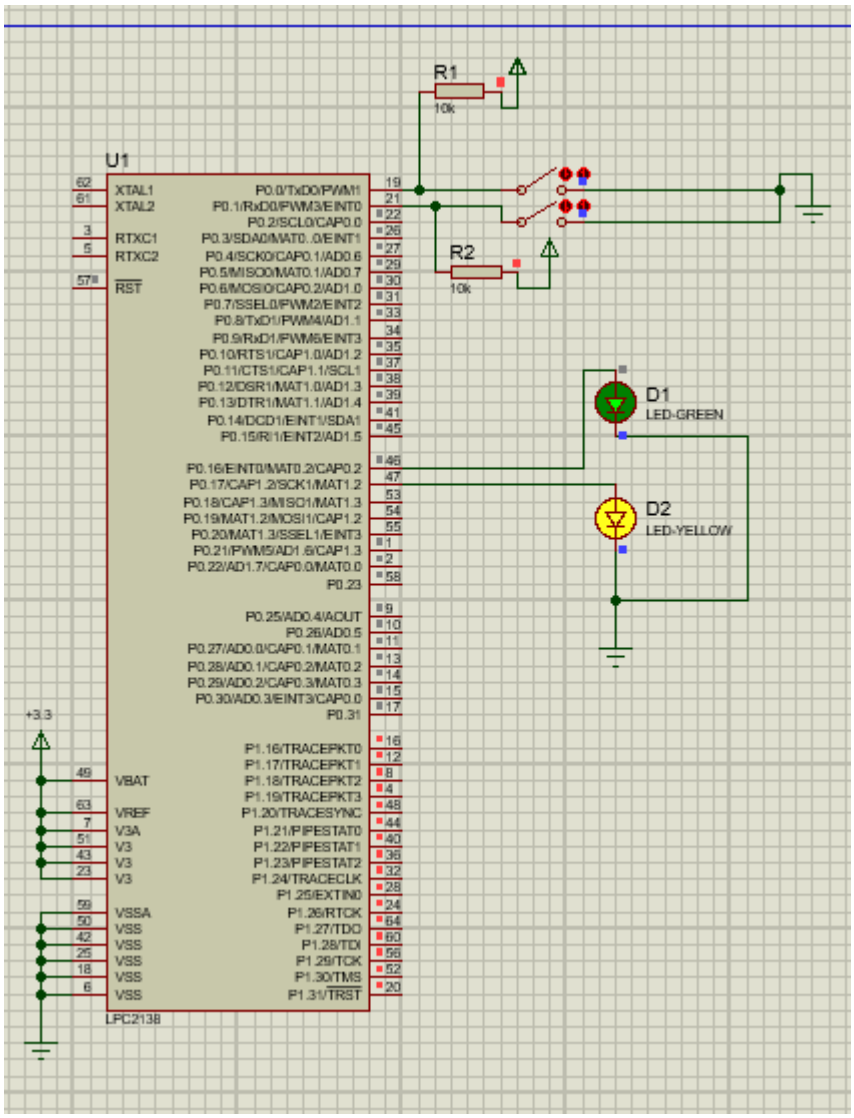

Registers
Register
Value
Current
R0 0x40000060
R1 0x40000060
R2 0x40000060
R3 0x40000060
R4 0x00000000
R5 0x40000000
R6 0x00000000
R7 0x00000000
R8 0x00000000
R9 0x00000000
R10 0x0000023C
R11 0x00000000
R12 0x0000019C
R13 (SP) 0x40000460
R14 (LR) 0x000000C0
R15 (PC) 0x0000019C
CPSR 0x600000D3
SPSR 0x00000000
User/System
Fast Interrupt
Interrupt
Supervisor
Abort
Undefined
Internal
PC $ 0x0000019C
Mode Supervisor
States 866
Sec 0.00007217
Disassembly
0x00000198 E12FFF1E BX R14
11: IODIR0 |= (LED1 | LED2);
12:
13: // Configure switches as input
0x0000019C E59F0084 LDR R0,[PC,#0x0084]
0x000001A0 E5900008 LDR R0,[R0,#0x0008]
Project
Project: lab2
Target 1
Source Group 1
Startup.s
labb2.c
labb2.c
1 //CH.SC.U4CSE24109
2 #include <LPC21xx.h>
3 #define SW1 (1<<0) // Switch at P0.0
4 #define SW2 (1<<1) // Switch at P0.1
5 #define LED1 (1<<16) // LED1 at P0.16
6 #define LED2 (1<<17) // LED2 at P0.17
7
8 int main(void)
9 {
10 // Configure LEDs as output
11 IODIR0 |= (LED1 | LED2);
12
13 // Configure switches as input
14 IODIR0 &= ~(SW1 | SW2);
15
16 while(1)
17 {
18 // Switch 1
19 if(IOPIN0 & SW1)
20 IOSET0 = LED1;
21 else
22 IOCLR0 = LED1;
23
24 // Switch 2
25 if(IOPIN0 & SW2)
26 IOSET0 = LED2;
27 else
28 IOCLR0 = LED2;
29 }
30 }
Command
Running with Code Size Limit: 32K
Load "C:\\Users\\ch.sc.u4cse24109\\Desktop\\CH.SC.U4CSE24109\\Lab-2\\Objects\\lab2.axf"
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 572 Bytes (1%)
ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate
Call Stack + Locals
Name
Location/V...
Type
main 0x0000019C int f()
Call Stack + Locals
Memory 1
Real-Time Agent: Target Reset

OUTPUT:


R1
10k
R2
10k
U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
P0.6/MOSI0/CAP0.2/AD1.0
P0.7/SSEL0/PWM2/EINT2
P0.8/TxD1/PWM4/AD1.1
P0.9/RxD1/PWM6/EINT3
P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/ACUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
P0.31
P1.16/TRACEPKT0
P1.17/TRACEPKT1
P1.18/TRACEPKT2
P1.19/TRACEPKT3
P1.20/TRACESYNC
P1.21/PIPESTAT0
P1.22/PIPESTAT1
P1.23/PIPESTAT2
P1.24/TRACECLK
P1.25/EXTIN0
P1.26/RTCK
P1.27/TDO
P1.28/TDI
P1.29/TCK
P1.30/TMS
P1.31/TRST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
+3.3
LPC2138
D1
LED-GREEN
D2
LED-YELLOW


R1
10k
R2
10k
U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
P0.6/MOSI0/CAP0.2/AD1.0
P0.7/SSEL0/PWM2/EINT2
P0.8/TxD1/PWM4/AD1.1
P0.9/RxD1/PWM6/EINT3
P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/ACUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
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P1.24/TRACECLK
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P1.26/RTCK
P1.27/TDO
P1.28/TDI
P1.29/TCK
P1.30/TMS
P1.31/TRST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
+3.3
LPC2138
D1
LED-GREEN
D2
LED-YELLOW


R1
10k
R2
10k
U1
XTAL1
XTAL2
RTXC1
RTXC2
RST
VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
VSS
VSS
VSS
VSS
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
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P1.31/TRST
LPC2138
D1
LED-GREEN
D2
LED-YELLOW


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RTXC1
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RST
P0.0/TxD0/PWM1
P0.1/RxD0/PWM3/EINT0
P0.2/SCL0/CAP0.0
P0.3/SDA0/MAT0..0/EINT1
P0.4/SCK0/CAP0.1/AD0.6
P0.5/MISO0/MAT0.1/AD0.7
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P0.7/SSEL0/PWM2/EINT2
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P0.10/RTS1/CAP1.0/AD1.2
P0.11/CTS1/CAP1.1/SCL1
P0.12/DSR1/MAT1.0/AD1.3
P0.13/DTR1/MAT1.1/AD1.4
P0.14/DCD1/EINT1/SDA1
P0.15/RI1/EINT2/AD1.5
P0.16/EINT0/MAT0.2/CAP0.2
P0.17/CAP1.2/SCK1/MAT1.2
P0.18/CAP1.3/MISO1/MAT1.3
P0.19/MAT1.2/MOSI1/CAP1.2
P0.20/MAT1.3/SSEL1/EINT3
P0.21/PWM5/AD1.6/CAP1.3
P0.22/AD1.7/CAP0.0/MAT0.0
P0.23
P0.25/AD0.4/ACOUT
P0.26/AD0.5
P0.27/AD0.0/CAP0.1/MAT0.1
P0.28/AD0.1/CAP0.2/MAT0.2
P0.29/AD0.2/CAP0.3/MAT0.3
P0.30/AD0.3/EINT3/CAP0.0
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VBAT
VREF
V3A
V3
V3
V3
VSSA
VSS
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D1
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